D3.3 Homeostasis

Notes written for this site. Slides, videos, worksheets and self-check questions: whatthecell.org/subtopics/sl-d3-3.html

Written for this site from the course textbook and class slides. Use the page numbers to read the book's fuller account. Image credits are on the last slide of the slides for this subtopic.

THE PRINCIPLE

D3.3.1 Homeostasis: keeping the internal environment steady

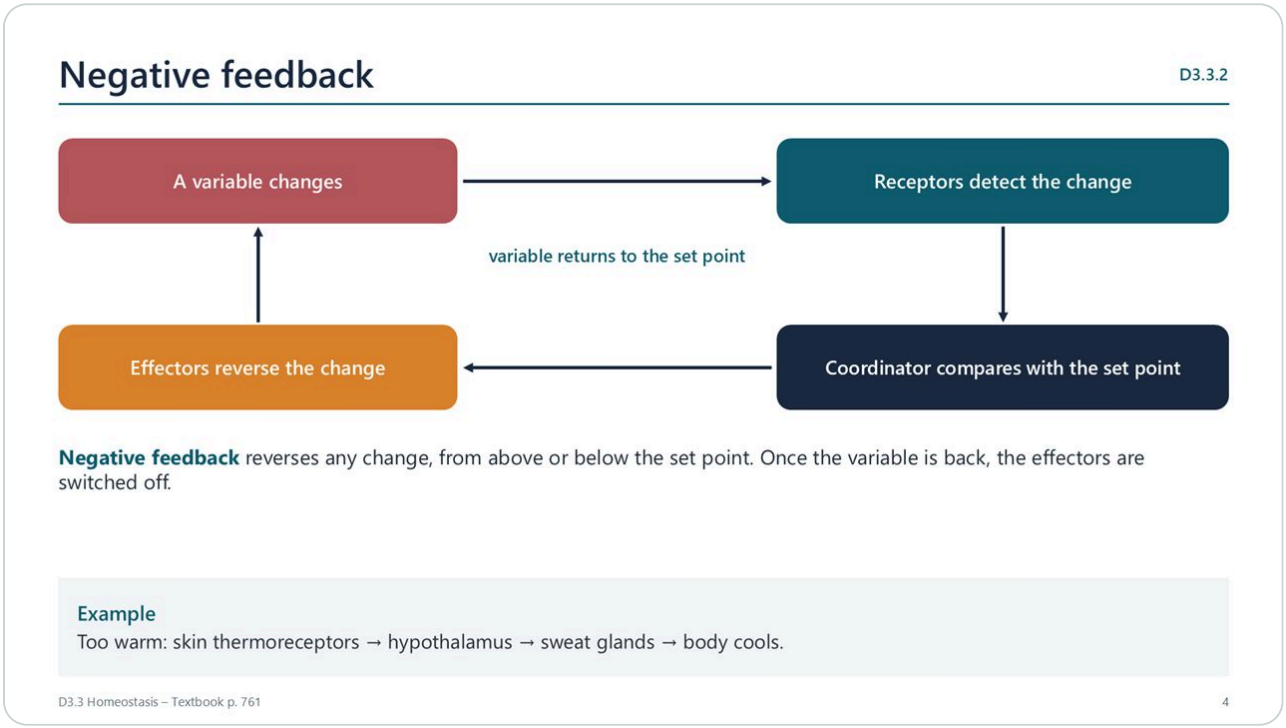
p. 760

Homeostasis is maintaining the internal environment within preset limits, despite changes in the external environment. In humans, the controlled variables include **body temperature**, **blood pH**, **blood glucose concentration** and **blood osmotic concentration**.

Watch out: homeostasis is about the internal environment. It does not mean moving to somewhere more comfortable.

D3.3.2 How negative feedback loops work

p. 761



A receptor detects a change, a coordinator compares it with the set point, and an effector acts in the opposite direction. **Negative feedback** returns a variable to the set point from above and from below. Positive feedback would move it further away, which is why homeostasis does not use it.

D3.3.3 How hormones regulate blood glucose

pp. 761–763

Blood glucose is normally about 90 mg per 100 cm^3. The brain has no glycogen store, so it depends on a steady supply from the blood.

	GLUCOSE TOO HIGH	GLUCOSE TOO LOW
DETECTED BY	Beta cells in the islets of Langerhans	Alpha cells in the islets of Langerhans
HORMONE	Insulin	Glucagon
TARGET CELLS	Liver, muscle and other tissues	Liver
EFFECT	Glucose uptake; storage as glycogen; more used in respiration	Glycogen broken down; glucose released

Watch out: glucagon is a hormone, and glycogen is the storage product. The pancreas itself detects the change, so the hypothalamus is not involved.

D3.3.4 The changes behind type 1 and type 2 diabetes

pp. 764–765

	TYPE 1	TYPE 2
PHYSIOLOGICAL CHANGE	The immune system destroys beta cells, so little or no insulin is produced	Insulin is produced, but target cells do not respond well to it
RISK FACTORS	Genes, and possibly exposure to some viruses	Overweight, inactivity, age, family history
PREVENTION	None known at present	Healthy diet, regular exercise, healthy weight
TREATMENT	Insulin, with regular monitoring of blood glucose	Diet and exercise, sometimes medicine

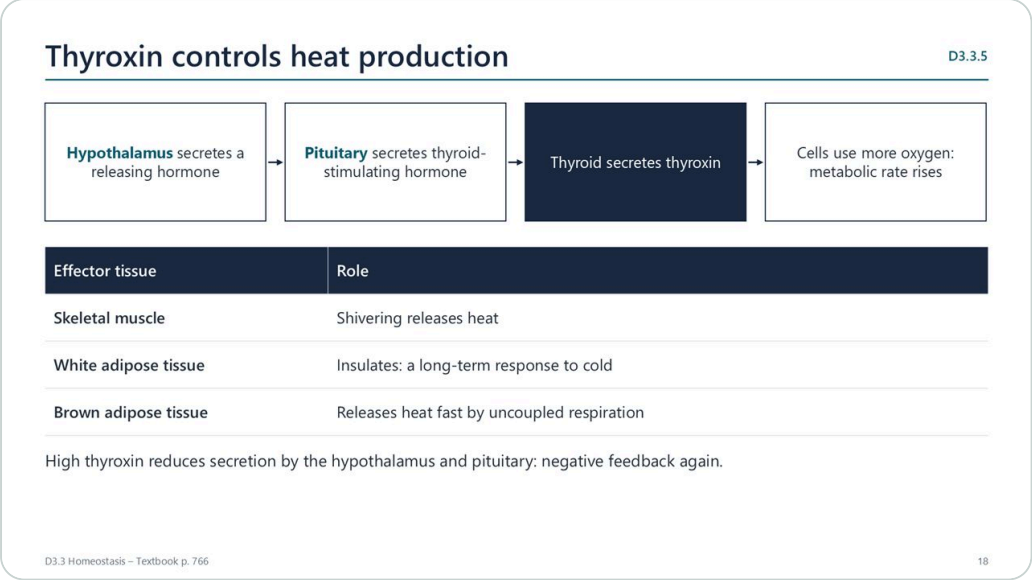
In either type, blood glucose stays raised and glucose may appear in urine. Long-term effects can include damage to the kidneys, eyes, nerves and blood vessels.

Watch out: do not say that type 2 is simply caused by a high-sugar diet. Such a diet is linked with it, and is one of several risk factors.

D3.3.5 Temperature control as negative feedback

p. 766

Peripheral thermoreceptors in the skin detect temperature changes. The **hypothalamus** compares these with the set point and signals to effectors, including muscle and adipose tissue. Hormones are involved too. The hypothalamus secretes a releasing hormone, the **pituitary gland** secretes thyroid-stimulating hormone, and the thyroid gland secretes **thyroxin**. Thyroxin raises oxygen consumption and metabolic rate, which generates heat.



D3.3.6 How humans regulate body temperature

pp. 766–767

	TOO HOT	TOO COLD
SKIN ARTERIOLES	Vasodilation: more blood to the skin	Vasoconstriction: less blood to the skin
SWEAT GLANDS	More sweat, which evaporates	Less sweat
HAIR	Muscles relax; hairs lie flat	Muscles contract; hair erection traps air
MUSCLE	No shivering	Shivering generates heat
BROWN ADIPOSE TISSUE	Little activity	Uncoupled respiration releases heat

Sweat cools the skin because evaporation breaks hydrogen bonds, which takes heat energy.

Uncoupled respiration in brown adipose tissue releases energy as heat rather than storing it in ATP, which is important for newborn babies. Birds and mammals also use behaviour to regulate temperature.

Watch out: blood vessels do not move up or down in the skin. Arterioles widen or narrow, which changes how much blood flows through the surface capillaries.

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